LaTex - Overleaf Guide.

Shortly, what is LaTeX?

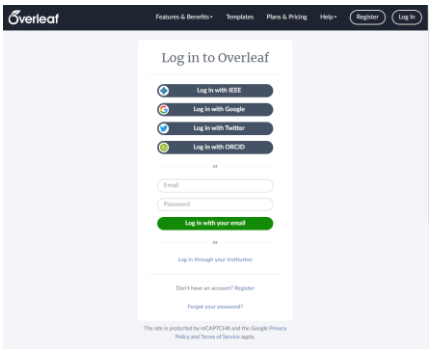
LaTeX is a typesetting system which includes features designed for the production of technical and scientific documentation. LaTeX is the de facto standard for the communication and publication of scientific documents.

To use LaTeX, we can find different software (you can google it), however, I advise you to use an online platform called Overleaf. It has many advantages, and here are a few of them:

1. The possibility of teamwork.
2. Version control.
3. Access from different devices, etc.

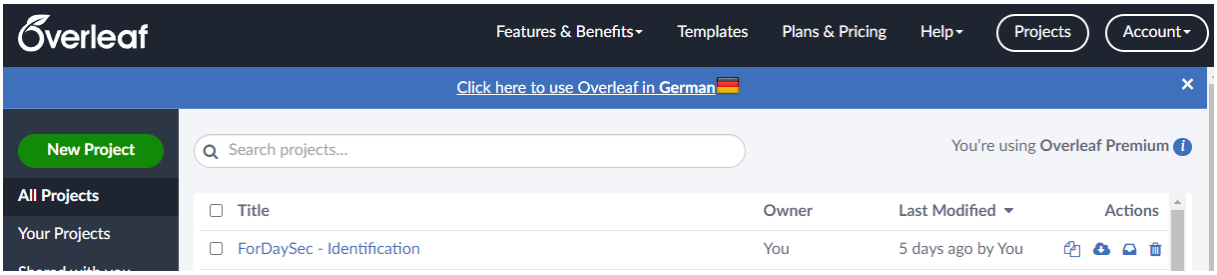
Overleaf step-by-step

Step 1. Go to https://www.overleaf.com/ and you will see such page:



Step 2. If you already have an account, you can press "Log in ..." using the appropriate option. If you do not have an account, press "Register" and do it.

Step 3. After registering, you will see a page with your projects and Overleaf information.



Step 4. In Overleaf you can create your own document or use existing templates for different purposes. The goal of our seminar is to produce a scientific paper. Usually, for scientific journals, the IEEE template is used.

IEEE Advancing Technology for Humanity

Check https://www.ieee.org/ to find out more about IEEE.

Paper Template.

So, to get the template which we need, you should go to this link:
https://www.ieee.org/conferences/publishing/templates.html
Then, you will see different formats of the template.

Microsoft Word

- US letter (DOC, 30 KB) **Updated Jan 2019**
- A4 (DOC, 30 KB) **Updated Jan 2019**

LaTeX Template Instructions (PDF, 63 KB) [Be sure to use the template's **conference** mode.]

- Template (ZIP, 700 KB) **Updated October 2019**
- LaTeX Bibliography Files (ZIP, 309 KB)

Overleaf

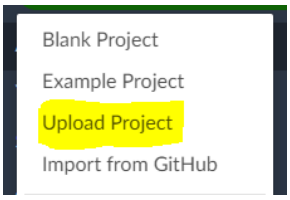
- When working in Overleaf, the template is available at https://www.overleaf.com/gallery/tagged/ieee-official

- Microsoft Word – we do not need it.
- LaTex Template Instructions - download it and read, because this document contains the main useful commands for your future paper.

Finally, we have two options to upload template to our Overleaf account.

I. Manual

1. Download the Template (ZIP).
2. Download LaTeX Bibliography Files.
3. Press „New Project“ in your Overleaf.
4. Chose “Upload Project”.



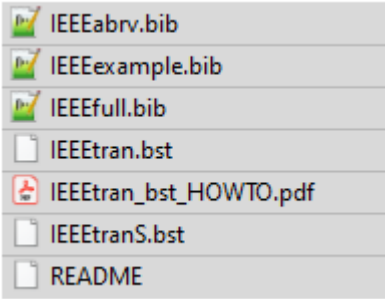
5. Select the Template (ZIP) file downloaded in (1).
6. Hopefully, your page will look like this:



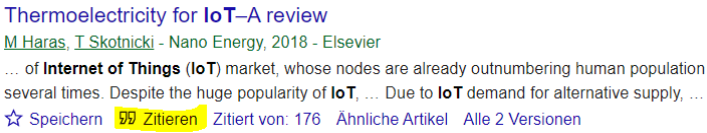
First field – The files which are contained in the project.
Middle field – The place, where you write code.
Right field – The preview zone. After changing the code, press “Recompile” and the Overleaf will show how it will look in PFD.

7. The important step is to install the bibliography. I think you have seen references at the end of every paper. In order to generate these references automatically, you need to use additional files (LaTeX Bibliography Files).

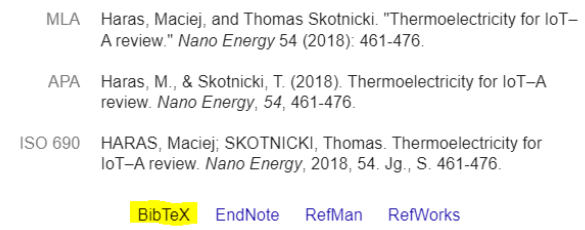
8. Extract files from the archive (LaTeX Bibliography Files).
9. In the top left-hand corner of Overleaf press Upload and select from your computer all extracted files.



10. In file IEEEexample.bib, there are examples of citations. Usually, for papers and books which you want to cite, the correct citations are created. You can find it on Google Scholar, ResearchGate, and similar resources.
Press "Cite".



Press "BibTeX".



Copy the text and insert it at the end of IEEEexample.bib file.

```
@article{haras2018thermoelectricity,
  title={Thermoelectricity for IoT--A review},
  author={Haras, Maciej and Skotnicki, Thomas},
  journal={Nano Energy},
  volume={54},
  pages={461--476},
  year={2018},
  publisher={Elsevier}
}
```

So, this reference will be available for citations. In order to cite it in the text, you need to use the id-name of the paper (in this case it is generated, but you can change it any time and use naming comfortable for you).

```
@article{haras2018thermoelectricity,
  title={Thermoelectricity for IoT--A review},
```

11. Then, return to the main file: conference_101719.tex. And delete everything from \section*{References} till \end{document}. (\end{document} – do not delete).

```
\section*{Acknowledgment}

The preferred spelling of the word ``acknowledgment'' in America is without
an ``e'' after the ``g''. Avoid the stilted expression ``one of us (R. B.
G.) thanks $\ldots$''. Instead, try ``R. B. G. thanks$\ldots$''. Put sponsor
acknowledgments in the unnumbered footnote on the first page.


\end{document}
```

12. Type here (before \end{document}) the following three lines:

```
\bibliographystyle{./IEEEtran}
\balance
\bibliography{./IEEEabrv,./IEEEexample}
```

13. And now, if you will cite something from IEEEexample.bib, it will be automatically added to references. Let's try to cite the paper, which we added in (10). Just type before lines from (12), this: \cite{haras2018thermoelectricity}.

```
\cite{haras2018thermoelectricity}

\bibliographystyle{./IEEEtran}
\balance
\bibliography{./IEEEabrv,./IEEEexample}
```

After pressing "Recompile" you will see in the preview zone: citation [1] and reference.

[1]

REFERENCES

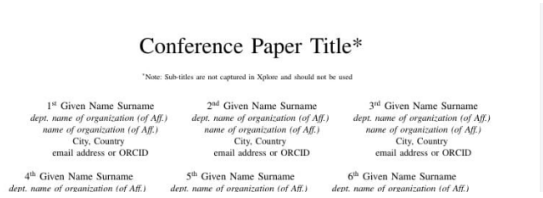
[1] M. Haras and T. Skotnicki, "Thermoelectricity for iot–a review," *Nano Energy*, vol. 54, pp. 461–476, 2018.

14. Done! Everything is now set up for full operation. You can find some additional LaTeX features and packages in Overleaf's manual or on Google. Good luck!

II. Automatic

1. Go to this site https://www.overleaf.com/gallery/tagged/ieee-official and find there IEEE Conference Template (Official). Press it.
2. You should see something like this:





Press "Open as Template". It will upload it to you Overleaf.

3. In order to add a bibliography, you need to go through steps 2, 7-14 from the Manual part.
4. Done! Good luck!